

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS (SUB-STRAND 2.2)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You are a scientist helping a farmer in Nakuru County. The farmer planted maize and beans on the same day after the long rains. Some seeds germinated within a few days, some took longer, and some healthy-looking seeds never sprouted. As the plants grew, they also differed in height, thickness, root depth and branching. Your task is to write a Claim-Evidence-Reasoning (CER) explanation that answers the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Use evidence from our unit: your seed drawings and dissections, the dormancy and germination investigations, the stem sections, the growth curves and the hormone notes. Write your answers in the space under each prompt. Use labelled drawings and arrows where they help. In Section 4 you will also revise your Day 1 model. Write in full sentences and use the scientific words we have learned.

Section 1: Seed structure and food supply

Compare a maize grain and a bean seed. (a) Name the main structures you saw in each, and say what each one does. (b) Describe how these two seeds differ in the number of cotyledons and in where the food is stored. (c) Explain how these differences could affect the way each seedling emerges and grows early on. Refer to your dissections of soaked seeds.

(a) Maize grain: one cotyledon (the scutellum), a large endosperm that stores food, a protective coleoptile around the shoot and a coleorrhiza around the root. Bean seed: a testa (seed coat) that protects the seed, a hilum (scar where it was attached), a micropyle (small pore for water entry), two large cotyledons that store food, a plumule (the future shoot) and a radicle (the future root). (b) Maize is a monocot with one cotyledon and its food is stored mainly in the endosperm. Bean is a dicot with two cotyledons, and its food is stored in the cotyledons. (c) The embryo uses the stored food for energy and building materials until the leaves can make food. Because the stores are held in different places, the two seedlings emerge differently: the maize shoot is protected by the coleoptile as it pushes up, while the bean seedling carries its food-storing cotyledons with it. So seed structure is one reason maize and beans did not behave the same way, even when planted on the same day after the same rain.

Section 2: Dormancy and germination conditions

Some of the farmer's seeds looked healthy but did not germinate, and others were late. Write a CER explanation using the two investigations we did. Claim: give two different reasons why seeds planted

Claim: Seeds can germinate late or fail because they are dormant, or because one of the conditions needed for germination is missing or the seed is not alive. Evidence: In the dormancy investigation, seeds that were rubbed with sandpaper, soaked in warm water or leached took up water and responded sooner than the untreated control seeds. In the germination sets, seeds with water, air and warmth germinated. Dry seeds did not germinate, seeds submerged with little oxygen did not germinate normally, cool seeds were slower, and boiled (non-viable) seeds did not germinate at all. Reasoning: A dormant seed is alive but rests even when conditions look suitable, for example because of a hard seed coat that blocks water, chemical inhibitors, or an embryo that is not ready. Scarifying, warm-water soaking and leaching break this block, which is why treated seeds responded sooner. Germination

after the same rain may germinate late or fail. Evidence: describe what you observed in the dormancy treatments (control, sandpaper, warm water, leaching or chilling) and in the germination sets (water, air and warmth; dry; little oxygen; cool; boiled seeds). Reasoning: explain why each result happened, using the words dormancy, viable, water, oxygen and temperature.	needs water to start the seed's chemical reactions, oxygen for respiration to release energy, and a suitable temperature. Boiled seeds are dead (not viable), so they cannot germinate whatever the conditions. In the farmer's field, waterlogged soil, dry patches, cool spots, dormant seeds or dead seeds could therefore each explain why some seeds were late or never sprouted.
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Section 3: How the seedlings emerge and keep growing

(a) Compare how maize and bean seedlings emerge from the soil. Use the terms hypogeal and epigeal. (b) Explain how primary growth and secondary growth differ, and where in the plant each one happens. (c) Use your stem-section observations to explain why a bean plant can become thicker and branch while maize grows mainly in length. (d) Describe the shape of the growth curve you plotted and what it tells you about growth over time.	(a) Maize shows hypogeal germination: the cotyledon stays below the ground and the shoot grows up. Bean shows epigeal germination: the cotyledons are lifted above the ground as the stem grows. (b) Primary growth is an increase in length. It happens at the apical meristems at the tips of roots and shoots, where cells divide and then elongate. Secondary growth is an increase in girth (thickness). It happens at lateral meristems, the vascular cambium and cork cambium, in dicots and woody plants such as Grevillea and Eucalyptus. (c) In the bean (dicot) stem, the vascular bundles were in a ring with cambium between the xylem and phloem, so the stem can thicken. In the maize (monocot) stem, the bundles were scattered and there was no cambium, so maize grows mainly in length. (d) The growth curve was S-shaped: a slow start (lag phase), a rapid growth phase, then a levelling off. This shows that the rate of growth changes over time and that the two crops can differ in how fast and how tall they grow even when planted on the same day. Maize grew faster and taller than beans in the same period.
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Section 4: Hormones and your final CER explanation

Write your final explanation to answer the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Claim: state your answer in one or two sentences. Evidence: give at least three pieces of evidence	Claim: The farmer's maize and beans differed because their seeds have different structures, because the seeds met different conditions (water, oxygen, warmth) and had different dormancy status, and because growth in the plants is driven by different meristems and controlled by hormones. Evidence: (1) Maize has one cotyledon and a large endosperm, while bean has two cotyledons, and they emerge differently (maize hypogeal, bean epigeal). (2) Seeds with water, air and warmth germinated, while dry, low-oxygen, cool or boiled seeds did not, and seeds with dormancy-breaking treatments responded sooner than the control. (3) The bean stem has cambium and can thicken and branch, while the maize stem has none and grows mainly in length, and both showed S-shaped growth curves. (4) A bean seedling with its tip removed grew more side branches but was shorter, while the intact seedling grew taller with few branches. Reasoning: Stored food and seed structure decide how the embryo is supplied and how the seedling emerges. Water, oxygen and suitable temperature allow the embryo to grow, and dormancy or dead seeds
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from different lessons, including at least one about hormones (for example, the bean seedling with its tip removed). Reasoning: link your evidence to your claim and explain how hormones work together with water, oxygen, temperature and seed structure. Finally, say how your Day 1 model or drawing has changed, and describe one thing you added.	explain why some healthy-looking seeds did not sprout. Once growing, cell division and elongation at apical meristems give length, and cambium gives thickness. Hormones coordinate all this: gibberellins and abscisic acid affect dormancy and germination time, auxin from the shoot tip causes apical dominance and shoot elongation, and cytokinins affect branching. Removing the tip removed the auxin, which released the side buds, so this supports the role of hormones. Growth is an increase in size, while development is the change in form and function as the seed becomes a mature plant. My Day 1 model only showed a seed and a shoot. I revised it to add the labelled seed structures, the conditions water, oxygen and warmth, dormancy, the meristems, and arrows for hormones.
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Section 5: Applying your understanding

A farmer in Kakamega or Nyeri wants to improve germination and rooting for a school tree nursery. Give two practical pieces of advice, and explain each one using ideas from this unit. Mention one natural rooting source you tested.	Advice 1: Check that the seeds are viable and plant them in moist but not waterlogged soil in a warm place. Seeds need water, oxygen and suitable warmth to germinate, and waterlogged soil removes oxygen. If seeds have a hard coat and seem dormant, scarify them with sandpaper or soak them in warm water to let water in, because dormancy can be broken this way. Advice 2: To help stem cuttings root, treat them with a natural rooting source such as bulb onion, Aloe or cinnamon. In our nursery test, cuttings treated this way rooted differently from the control. Auxins are plant hormones that encourage rooting, so these sources can help the cuttings form roots. Healthy tree seedlings can then be shared to benefit the community and help reduce global warming.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy	All ideas are correct: seed structure, dormancy, germination conditions, primary and secondary growth, and hormone roles, with correct use of terms such as monocot, dicot, hypogeal, epigeal and meristem.	Most ideas are correct, with one or two small errors or terms used loosely.	Several ideas are incorrect or confused, for example mixing up primary and secondary growth or seed types.
Use of evidence	Gives at least three specific pieces of evidence from different lessons (seed dissections, germination and dormancy sets, stem sections, growth curves, tip-removal bean seedling).	Gives two relevant pieces of evidence, or three that are general.	Gives little or no evidence, or evidence that is not from our investigations.
Reasoning	Clearly explains how each piece of evidence supports the claim and links seed structure, conditions, dormancy,	Explains some links between evidence and claim, but some factors are listed rather than explained.	Mostly lists facts without explaining how they answer the question.

	meristems and hormones together.		
Connection to the driving question and revised model	Directly answers why maize and beans planted together differ in germination, emergence and growth, and clearly shows how the Day 1 model was revised with specific additions.	Answers the question and mentions a model change with few details.	Does not clearly answer the question or does not describe any change to the model.
Clarity of communication	Well organised in claim, evidence and reasoning, with full sentences, clear labelled drawings and correct vocabulary.	Mostly clear, with some gaps in organisation or labelling.	Hard to follow, with missing structure or unlabelled drawings.

ATTRIBUTION AND LICENSING

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